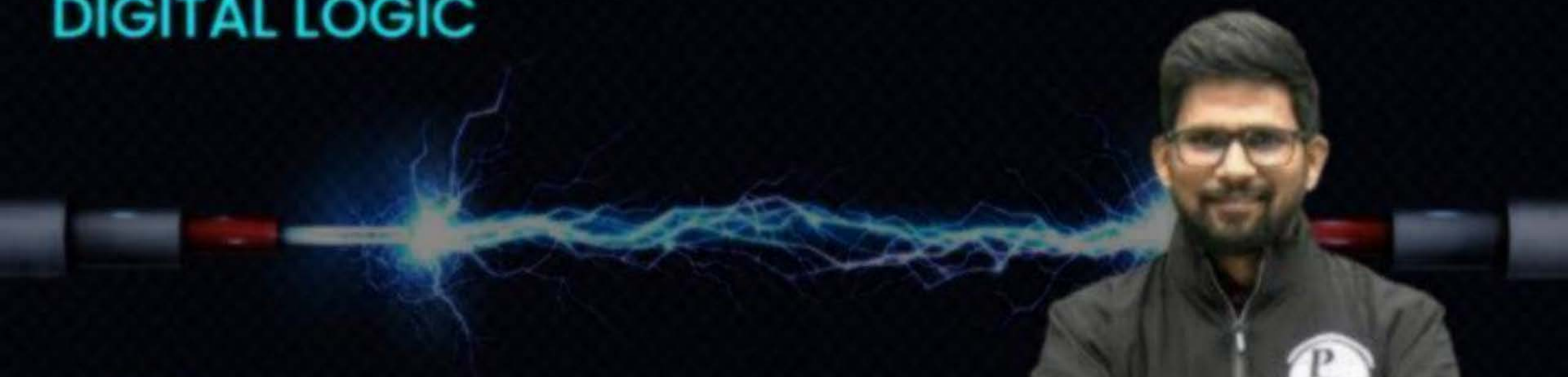
# COMPUTER SCIENCE & IT

## DIGITAL LOGIC



Lecture No. 11

## BOOLEAN THEOREMS AND GATES



By- Chandan Gupta Sir



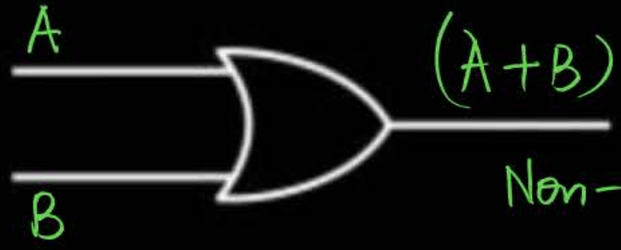

NOR gate



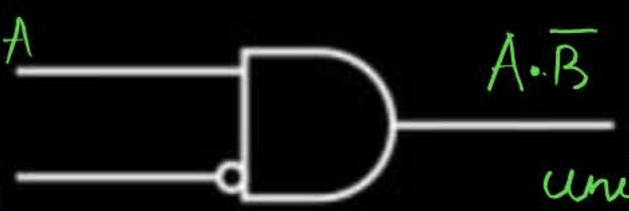

- gmp basics

## Question

Which of the following circuit can work as universal gate ?

(a) 

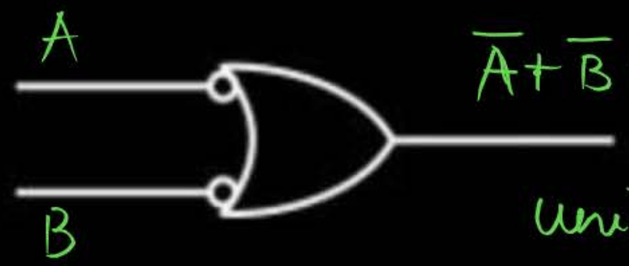
Non-universal

(b) 

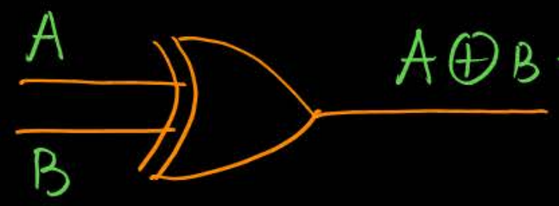
universal

(c) 

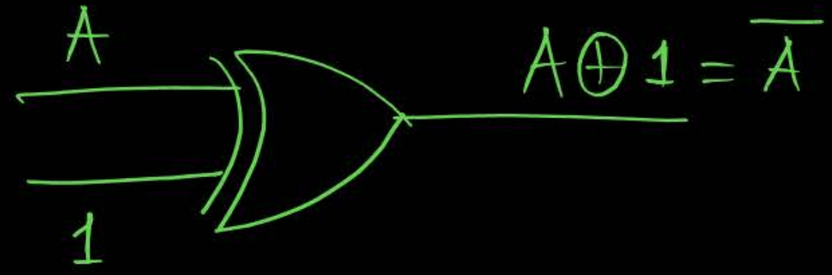
Non universal

(d) 

universal

- 


$A \oplus B = (\bar{A}B + A\bar{B})$ ⟶ non - universal




$A \oplus 1 = \bar{A}$

$A \oplus B \oplus AB = (A+B)$

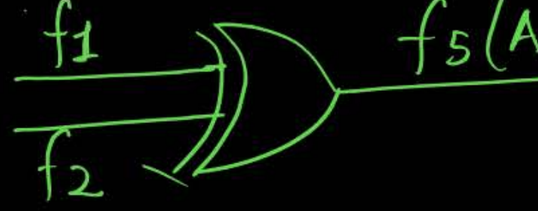
$A \oplus B \oplus (A+B) = A \cdot B$

$A \odot B \odot (A+B) = A \cdot B$

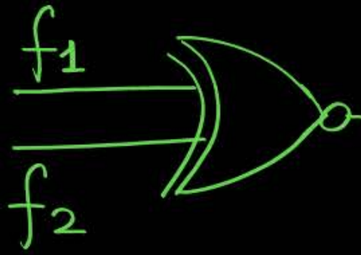
$A \odot B \odot AB = A+B$

- $f_1(A,B,C) = \Sigma(0,1,2,4)$

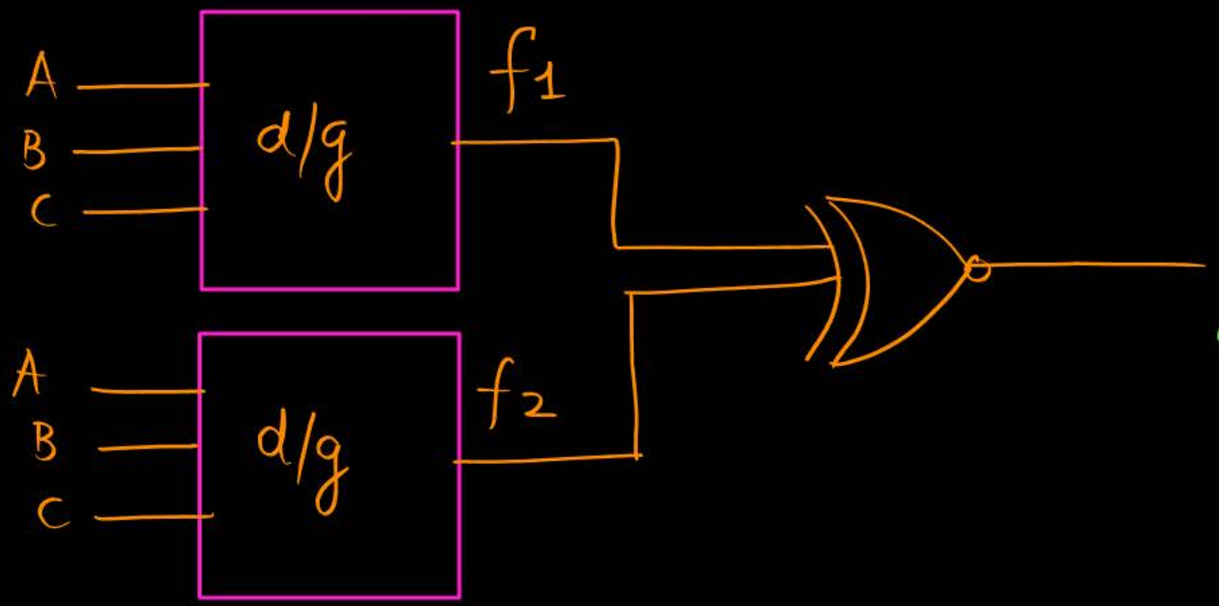
$f_2(A,B,C) = \Sigma(0,2,3,5)$

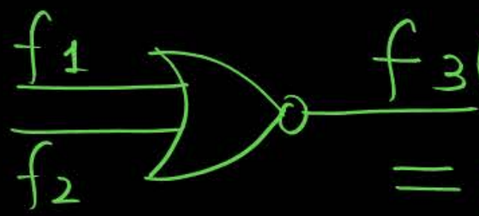



$f_5(A,B,C) = \Sigma(1,3,4,5)$

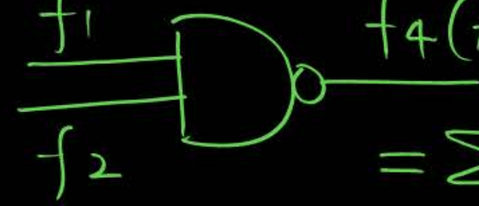



$f(A,B,C) = \Sigma(0,2,6,7) = \pi(1,3,4,5)$




- 

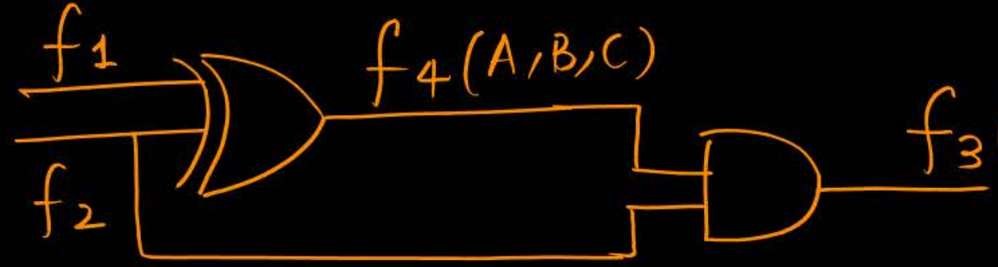

$f_3(A,B,C) = \Sigma(6,7)$

- 


$f_4(A,B,C) = \Sigma(1,3,4,5,6,7)$

- $f_1(A,B,C) = \Sigma(0,2,4,6)$

  $f_2(A,B,C) = \Sigma(1,2,4,5,7)$



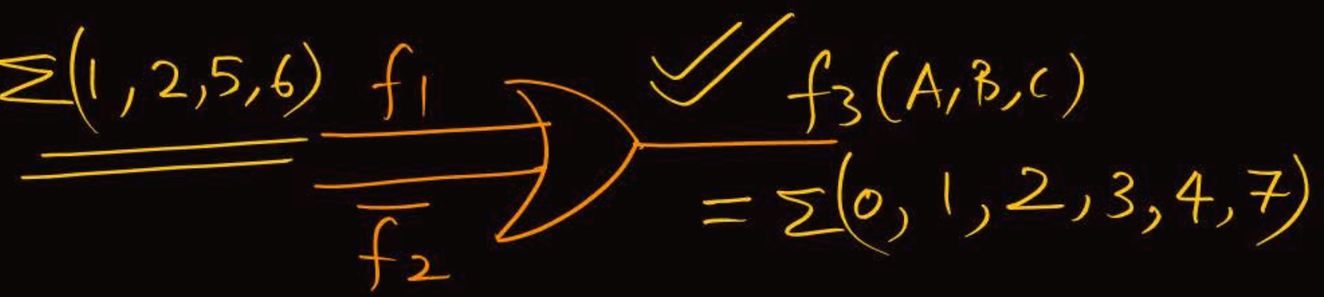

$f_3(A,B,C) = \Sigma(1,5,7)$

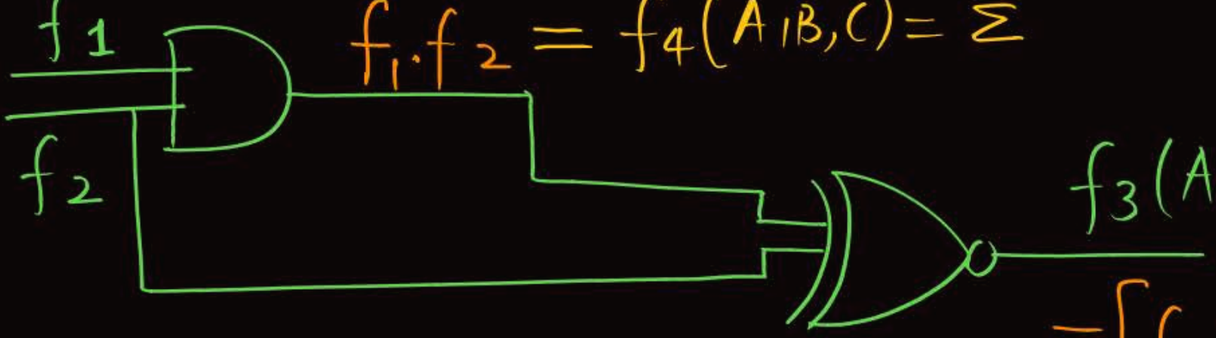
$f_4(A,B,C) = \Sigma(0,1,5,6,7)$

Q. $f_1(A,B,C) = \Sigma(1,2,4,7)$

$f_2(A,B,C) = \pi(0,3,4,7) = \Sigma(1,2,5,6)$

$\overline{f_2} = \Sigma(0,3,4,7)$




$f_3(A,B,C) = \Sigma(0,1,2,3,4,7)$

$= \overline{[f_1 f_2 \odot f_2]}$

$= \overline{f_1 f_2 \oplus f_2}$

$= \overline{f_2(f_1 \oplus 1)} = \overline{f_2 f_1} = (\overline{f_2} + f_1)$

$f_4(A,B,C) = \Sigma(1,2)$

$f_3(A,B,C) = \Sigma(0,1,2,3,4,7)$

# Concept of Duality :

→ To find out dual function of a given boolean function, we replace 'AND' operation by 'OR' operation and 'OR' operation by 'AND' operation while NOT remains as it is.

- $f(A,B,C) = AB + \bar{B}C = (B+C)(\bar{B}+A)$

  $f^D(A,B,C) = (A+B)\cdot(\bar{B}+C) = A\bar{B} + BC \neq f(A,B,C)$

- $[f^D(A,B,C)]^D = (A\cdot B) + \bar{B}\cdot C = f(A,B,C)$

$$f \xrightarrow{\text{dual}} f^D \xrightarrow{\text{dual}} [f^D]^D = f$$

- $f(A,B) = \overline{A}\cdot\overline{B} + A\overline{B} = \overline{B}(\overline{A}+A) = \overline{B}$

$$f^D(A,B) = (\overline{A}+\overline{B})\cdot(A+\overline{B}) = \overline{B} + A\cdot\overline{A} = \overline{B} = f(A,B)$$

↓ self dual boolean function.

$f(A,B,C) = AB + BC + AC = \Sigma(3,5,6,7)$

$f^D(A,B,C) = (A+B)\cdot(B+C)\cdot(A+C) = (B+AC)(A+C)$

$= AB + BC + AC$

$= f(A,B,C)$

↓ Self dual boolean function.

| A | B | |
|---|---|---|
| 1 | 1 | 0 → 6 |
| 1 | 1 | 1 → 7 |

| B | C | |
|---|---|---|
| 0 | 1 | 1 → 3 |
| 1 | 1 | 1 → 7 |

| A | C | |
|---|---|---|
| 1 | 0 | 1 → 5 |
| 1 | 1 | 1 → 7 |

$$f(A,B,C) = AB + BC + CA = \Sigma(3,5,6,7) = \Pi(0,1,2,4)$$

$$= \bar{A}\cdot B\cdot C + A\bar{B}C + AB\bar{C} + ABC$$

$$f^D(A,B,C) = (\bar{A}+B+C)\cdot(A+\bar{B}+C)(A+B+\bar{C})(A+B+C)$$
$$1\ 0\ 0 \qquad 0\ 1\ 0 \qquad 0\ 0\ 1 \qquad 0\ 0\ 0$$

$$= \Pi(0,1,2,4) = \Sigma(3,5,6,7) = f(A,B,C)$$

- $f_1(A,B,C) = \Sigma(1,2,3,4) = \Pi(0,5,6,7) = \bar{A}\bar{B}C + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}\bar{C}$

$$f_1^D(A,B,C) = (\bar{A}+\bar{B}+C)(\bar{A}+B+\bar{C})(\bar{A}+B+C)(A+\bar{B}+\bar{C}) = \Pi(3,4,5,6)$$
$$1\ 1\ 0 \qquad 1\ 0\ 1 \qquad 1\ 0\ 0 \qquad 0\ 1\ 1$$

$$\neq f_1(A;B,C)$$

$(0-7)$ $(0,7)$ $(1,6)$ $(2,5)$ $(3,4)$

$(0-15) \rightarrow$ $(0,15)$ $(1,14)$, $(2,13)$ $(3,12)$ $(4,11)$ $(5,10)$, $(6,9)$ $(7,8)$

$\Sigma 0$

$f \rightarrow \bar{A}\bar{B}\bar{C}\bar{D}\Sigma 0$

$\pi(15)$

$f^D \rightarrow (\bar{A}+\bar{B}+\bar{C}+\bar{D}) = \pi(15)$

1 1 1 1

- $f(A,B,C) = \Sigma(1,2,3,7)$ ⟶ self dual.
- $f_1(A,B,C) = \Sigma(0,2,4,6)$ ⟶ self dual.
- $f_2(A,B,C) = \Sigma(0,1,3,6)$ ⟶ Non-self dual
- $f_3(A,B,C) = \Sigma(0,1,2,4,6)$ ⟶ Non-self dual.

→ n-variables → $N = 2^n$ → combinations or terms $0 - (2^n - 1)$

How many boolean function we can make :

A, B → 2 variables $N = 2^2$ → combinations

| | | | |
|---|---|---|---|
| 0 | 00 | $\bar{A}\bar{B}$ | 4-terms or 4 Comb |
| 1 | 01 | $\bar{A}B$ | |
| 2 | 10 | $A\bar{B}$ | |
| 3 | 11 | $AB$ | |

$$f_1(A,B) = \Sigma(0) \qquad f_5(A,B) = \Sigma(0,1)$$
$$f_2(A,B) = \Sigma(1) \qquad f_6(A,B) = \Sigma(0,2)$$
$$f_3(A,B) = \Sigma(2)$$
$$f_4(A,B) = \Sigma 3$$

$$M = {}^{N}C_0 + {}^{N}C_1 + {}^{N}C_2 + {}^{N}C_3 + \cdots + {}^{N}C_N$$

$$M = 2^N = 2^{2^n}$$

N – combinations

$\frac{N}{2}$ groups $\Sigma$ ( ÷ ÷ ÷ – – – )

N/2 places

$2 \times 2 \times 2 \times \cdots$ N/2 times

$$S_D = 2^{N/2} = 2^{2^{n-1}}$$

→ n = 3 – variables ⟶ $N = 2^3$ comb or terms

no of boolean functions $M = 2^N = 2^8 = 256$

$S_D = 2^{2^2} = 2^4 = 16$

- $f(A,B) = \Sigma\, mil = \pi(0,1,2,3)$

$$= (A+B)(A+\overline{B})(\overline{A}+B)(\overline{A}+\overline{B})$$

$$= (A+0)(\overline{A}+0)$$

$$= A \cdot \overline{A}$$

$$= 0$$

$$f(A,B) = \Sigma(0,1,2,3) = \overline{A}\,\overline{B} + \overline{A}B + A\overline{B} + AB$$

$$= \overline{A} + A$$

$$= 1$$

H.W.

Q.1. $f(A,B,C,D) = \Sigma(0,1,2,3,7,9,10,11)$ self dual or NOT

Q.2. $f(A,B,C) = \Sigma(1,2,3,6)$ self dual or NOT

Q.3 $f(A,B,C) = AB \oplus BC \oplus AC$ ⟶ self dual or NOT

Q.4. $f(A,B,C,D) = \Sigma(1,2,3,7,9,10,11,15)$ self dual or NOT

Q.5. $f(A,B,C)$ ⟶ is self dual function

then $\overline{f}(A,B,C)$ will be self dual or NOT ?



# 2 Minute Summary

→ Imp basics.

Thank you

GW

*Soldiers !*

